unit 3 os

Explain the following concepts in detail.
i) Logical and Physical Address Space managed by memory.
ii) Dynamic Loading, Dynamic Linking and Shared Libraries

Sure! Here's a detailed explanation of the concepts you've asked about:

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i) Logical and Physical Address Space managed by Memory

Logical Address Space (LAS):

- A **logical address** (also known as a **virtual address**) is the address generated by the **CPU** during a program's execution.
- It is **not the actual address** of the data in the physical memory (RAM).
- The **Logical Address Space** refers to the set of all logical addresses generated by a program.
- Logical addresses are used because:
  - Programs are loaded into **different parts of memory** at different times.
  - They allow **isolation** between processes, improving security and reliability.
  - It allows **memory abstraction**, so the program doesn't need to worry about the actual location in RAM.

Physical Address Space (PAS):

- A **physical address** is the **actual location in RAM**.
- The **Physical Address Space** is the set of all physical addresses corresponding to the memory cells.
- Physical addresses are **computed by the Memory Management Unit (MMU)** using mapping from logical to physical addresses.

Address Translation:

- Done by the **Memory Management Unit (MMU)**.
- Example using **base + offset**:

- Logical address = 1000
- Base address in physical memory = 4000
- Physical address = base + logical = **5000**

Key Differences:

Feature	Logical Address	Physical Address
Generated by	CPU	MMU
User accessibility	Visible to user/program	Hidden from user
Binding time	Compile/Load/Execution time	Always at execution time
Use	Helps in memory abstraction	Actual memory used by process

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ii) Dynamic Loading, Dynamic Linking, and Shared Libraries

Dynamic Loading:

- In **dynamic loading**, a routine (like a function or module) is **not loaded until it is called**.
- This helps save memory because only the necessary modules are loaded during execution.
- Useful in large programs where not all code is needed all the time.
- Example:
  - A program has several features (say, image processing, text editing). If a user uses only the text feature, the image module isn't loaded.

Benefits:

- Reduces **initial load time**.
- Saves **RAM space**.
- Efficient **resource usage**.

Dynamic Linking:

- In **dynamic linking**, linking of some libraries or modules is **postponed until runtime**.
- Instead of copying the code of library functions into the program, only a **reference** (stub) is included during compilation.
- The actual linking to the shared library happens **when the program runs**.

Example:

- A C program using `math.h` doesn't include all math functions in the binary. Instead, at runtime, it links dynamically to `libm.so`.

Benefits:

- Reduces **executable size**.
- Enables **sharing of libraries** among different programs.
- Easier **updates**: updating the library updates all dependent programs.

Shared Libraries:

- A **shared library** is a file (like `.dll`, `.so`, `.dylib`) that can be used by **multiple programs simultaneously**.
- Instead of having a copy of the library code in every program, the OS loads **one copy into memory**, and multiple programs can use it.

Example:

- Windows `.dll` (Dynamic Link Library), Linux `.so` (Shared Object), macOS `.dylib`.

Benefits:

- Saves **disk space** and **memory**.
- Allows **modularity**: update/replace library independently.
- Facilitates **version control** and **security** (by isolating shared components).

Summary Table:

Concept	Purpose	When it Happens	Key Benefit
Dynamic Loading	Load code only when needed	Runtime	Reduces memory usage
Dynamic Linking	Link libraries at runtime	Runtime	Smaller binaries, easy updates
Shared Libraries	Share library code among apps	Runtime	Saves RAM/disk, easier maintenance

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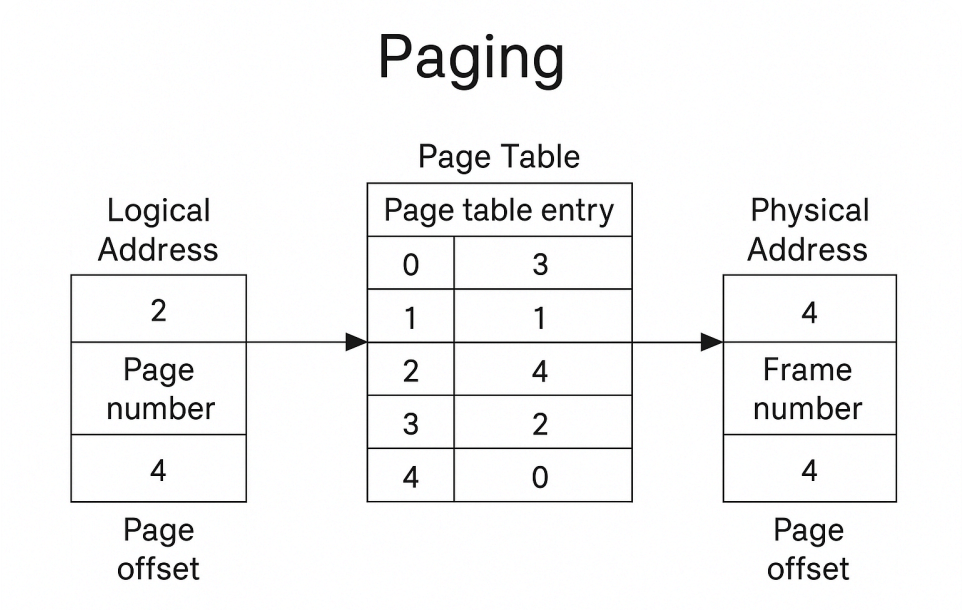
Let me know if you'd like diagrams or examples in code to further clarify!

Create image Outline the concepts of paging with suitable example.



Create image Outline the concepts of paging with suitable example.

Outline the concepts of paging with suitable example.

Sure! Here's an **outline of the concept of paging** in operating systems, along with a **simple example** to help you understand it better.

🧠 Concept of Paging

Paging is a **memory management technique** used by operating systems to manage how processes are stored in memory. It allows the **logical address space** (used by a program) to be **non-contiguous** in the **physical memory**, which avoids fragmentation and increases memory utilization.

🔹 Key Terms

Term	Description
Page	Fixed-size block of **logical memory**

Term	Description
Frame	Fixed-size block of **physical memory**
Page Table	A data structure used to **map logical pages to physical frames**
Logical Address	Address generated by CPU (page number + offset)
Physical Address	Actual address in RAM (frame number + offset)

🛠️ How Paging Works

1. The **logical address** is divided into:
   - **Page Number (p)**
   - **Offset (d)**
2. The **Page Number** is used to index the **Page Table**.
3. The **Page Table** provides the corresponding **Frame Number** in physical memory.
4. The **Physical Address** is calculated as:

$$\text{Physical Address} = \text{Frame Number} \times \text{Frame Size} + \text{Offset}$$

📘 Example

Let's say:

- Page Size = Frame Size = 1 KB = 1024 bytes
- Logical Address = 2050
- This gives:
  - Page Number (p) = 2050 / 1024 = 2 (integer part)
  - Offset (d) = 2050 % 1024 = 2

Assume:

- Page Table says **Page 2 is stored in Frame 5**

Then:

- Frame Number = 5
- Physical Address = 5 × 1024 + 2 = **5122**

✅ Advantages of Paging

- Eliminates external fragmentation.
- Allows efficient memory use.
- Enables use of **virtual memory** (swap between RAM and disk).

Would you like a printable version or a chart-style breakdown of this for revision purposes?

Explain briefly about different structures of page table.

Sure! Here's a **brief explanation of different structures of page tables** used in memory management:

📋 Different Structures of Page Table

1. Single-Level Page Table

- The simplest structure.
- One page table per process.
- Each entry maps one **page** to a **frame**.
- All entries are stored in **main memory**.

🔹 **Drawback:**

- Wastes memory if the process uses only a small portion of the address space.

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2. Multi-Level Page Table

- The page table is **broken into multiple levels** (e.g., two-level, three-level).
- The **first-level page table** points to second-level page tables, and so on.
- Reduces memory usage by **loading only the needed parts** of the table.

🔹 **Example:**

- Two-level: [Page Number] is split into:
  - First 10 bits → index into first-level page table.
  - Next 10 bits → index into second-level page table.

🔹 **Advantage:**

- Saves memory in sparse address spaces.

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3. Inverted Page Table

- One entry for **each frame in physical memory**, not for each page.
- Stores: (Process ID, Page Number, Frame Number)
- Only **one global page table** for all processes.

🔹 **Advantage:**

- Very **memory-efficient** for large address spaces.

🔹 **Drawback:**

- Lookup can be slower, often requires a **hash table** for faster access.

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4. Hashed Page Table

- Used for handling **large address spaces**.
- A **hash function** maps the page number to a hash table index.
- Each entry has a **linked list** of (Page Number, Frame Number) pairs.

🔹 **Advantage:**

- Efficient in handling **sparse and large** virtual address spaces.

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5. Segmented Page Table

- Combines **segmentation and paging**.
- Each **segment** has its own **page table**.
- Provides **logical grouping** and fine control over memory.

🔹 **Used in:**

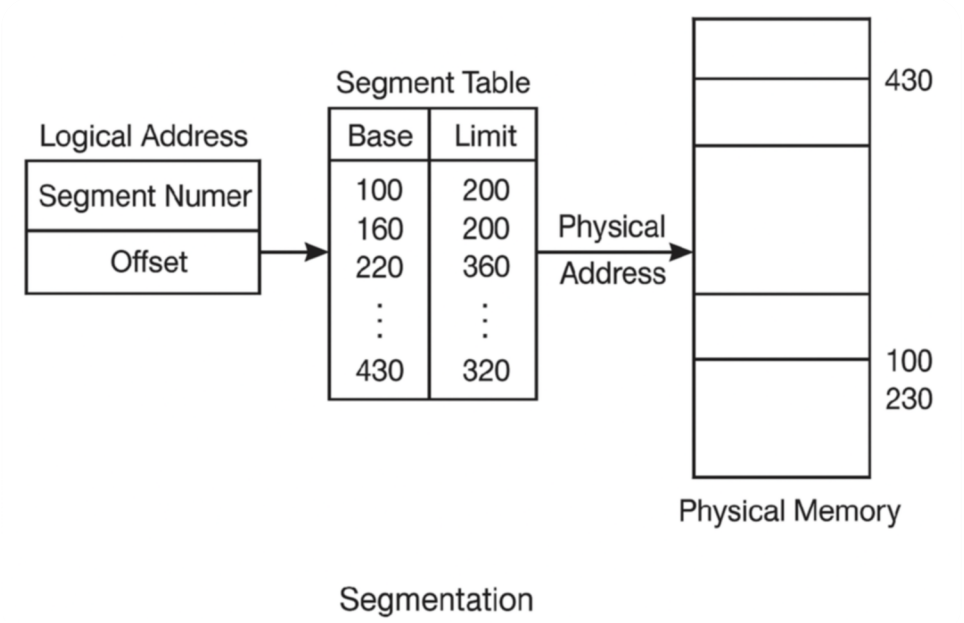
- Systems where both **protection and dynamic memory allocation** are important.

✅ Summary Table

Structure	Memory Efficient	Lookup Speed	Use Case
Single-Level	❌	✅	Simple systems
Multi-Level	✅	⚠️ Medium	Large address spaces
Inverted	✅✅	❌	OS with many small processes
Hashed	✅	✅	64-bit systems
Segmented Page Table	⚠️ Medium	✅	Needs logical memory grouping

Let me know if you'd like a visual diagram for these structures!

Draw the diagram of segmentation memory management scheme
and explain its principle.



segmentation memory management scheme
and explain its principle.

📘 Segmentation Memory Management Scheme

🧠 What is Segmentation?

Segmentation is a **memory management technique** where the **logical memory is divided into variable-sized segments**, based on the **logical divisions of a program** such as:

- Code segment
- Data segment
- Stack segment
- Heap segment

Each segment is treated as a **separate entity**, which reflects the **programmer's view** of memory.

⚙️ Principle of Segmentation

1. **Logical Address Structure:**
   - A logical address is divided into two parts:
     - **Segment Number (s)** – Identifies the segment.
     - **Offset (d)** – Distance from the start of the segment.
2. **Segment Table:**
   - Each process has a **segment table**.
   - Each entry contains:
     - **Base** – Starting physical address of the segment.
     - **Limit** – Length of the segment.
3. **Address Translation:**
   - When a program accesses a memory location, the CPU generates a logical address (s, d).
   - The system:
     - Looks up **segment s** in the segment table.
     - Checks if **d < limit** (to prevent illegal access).
     - If valid, **physical address = base + offset (d)**.

📦 Example

- Segment table for a process:

Segment	Base	Limit
0	1000	400
1	1400	600
2	2000	500

- Logical Address: (1, 200)
  - Segment = 1
  - Offset = 200
  - Base = 1400
  - So, Physical Address = 1400 + 200 = **1600**

✅ Advantages

- Reflects **logical structure** of programs.
- Enables **modular programming**.
- Provides **memory protection** between segments.
- Facilitates **dynamic memory allocation**.

❌ Disadvantages

- Can cause **external fragmentation**.
- Requires **complex memory management**.
- Slower than paging in some systems.

Let me know if you'd like this in a note-friendly or printable format!

Discuss about Virtual memory and Demand paging.

Sure! Here's a clear and concise explanation of **Virtual Memory** and **Demand Paging** — two fundamental concepts in modern operating systems:

🧠 Virtual Memory

🔹 Definition:

Virtual memory is a **memory management technique** that gives an application the **illusion of having a large, continuous block of memory**, even if the actual physical memory (RAM) is smaller.

🔹 Key Concepts:

- Virtual memory allows **processes to run even if not completely in RAM**.
- It uses both **main memory (RAM)** and **secondary storage (like a hard disk)**.
- Implemented through **paging** or **segmentation**.
- Managed by the **Operating System (OS)** with the help of hardware like the **Memory Management Unit (MMU)**.

🔹 Benefits:

- **Larger address space** than physical RAM.
- **Isolation and protection** between processes.
- Efficient **multitasking**.
- Allows **lazy loading** (only load parts of programs that are actually used).

📦 Demand Paging

🔹 Definition:

Demand paging is a **virtual memory mechanism** where pages of a program are **loaded into memory only when they are needed** (i.e., on demand).

🔹 How It Works:

1. Initially, **no pages** (or a few) are loaded into memory.
2. When the program tries to access a page that's not in memory:
   - A **page fault** occurs.
   - The OS loads the required page from disk to RAM.
   - The instruction is **re-executed**.

🔹 Key Components:

- **Page Table** – Keeps track of whether a page is in memory or on disk.
- **Page Fault Handler** – Handles the process of fetching a page from secondary memory.

🔹 Advantages:

- Saves **RAM space** by loading only what's necessary.
- Improves **system responsiveness** and **performance** with smart page replacement.
- Enables **execution of large programs** in limited memory.

🔹 Disadvantages:

- **Page faults** cause delays (slow disk I/O).
- Excessive page faults can lead to **thrashing** (system slows down heavily).

🧾 Comparison Table

Feature	Virtual Memory	Demand Paging
Concept	Abstracts memory beyond physical RAM	Technique to load pages only when needed
Purpose	Run large processes with limited RAM	Reduce memory usage by lazy loading

Feature	Virtual Memory	Demand Paging
Page Fault	May occur if accessed page not in RAM	Central mechanism
Performance Impact	Improves multitasking	Efficient but can be slow if mismanaged

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Let me know if you'd like a diagram or memory flow chart to help visualize how these work!

Explain the various page replacement strategies.

Sure! Here's a clear and concise explanation of **various page replacement strategies**, which are used in virtual memory systems when a page needs to be loaded into memory, but no free frame is available.

🧠 What is Page Replacement?

When the memory is full and a **page fault** occurs (i.e., the required page is not in RAM), the OS must **replace an existing page** with the new one. The strategy it uses to decide **which page to remove** is called a **page replacement algorithm**.

🔄 Common Page Replacement Strategies

1. FIFO (First-In, First-Out)

- Oldest page (loaded first) is replaced.
- Implemented using a **queue**.

✅ **Pros:**

- Simple to implement

❌ **Cons:**

- May replace frequently used pages
- Suffers from **Belady's anomaly** (more frames can lead to more page faults)

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2. LRU (Least Recently Used)

- Replaces the page that **has not been used for the longest time**.

✅ **Pros:**

- Approximates optimal performance
- Good for real-world programs

❌ **Cons:**

- Requires **tracking usage history** (can be complex)
- More overhead than FIFO

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3. Optimal Page Replacement

- Replaces the page that **will not be used for the longest time in the future**.

✅ **Pros:**

- Gives **minimum number of page faults**

❌ **Cons:**

- **Not possible in practice**, since future references are unknown
- Used for benchmarking only

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4. Clock (Second-Chance) Algorithm

- Circular version of FIFO.

- Each page has a **reference bit**:
  - If bit is 0 → replace the page.
  - If bit is 1 → give it a "second chance" and move on.

✅ **Pros:**

- Efficient approximation of LRU
- Less overhead than LRU

❌ **Cons:**

- Not as accurate as pure LRU

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5. NRU (Not Recently Used)

- Pages are classified into 4 classes based on **reference** and **modify** bits.
- Pages with lowest class are replaced first.

✅ **Pros:**

- Considered both usage and modification
- Good balance between performance and cost

❌ **Cons:**

- Needs hardware support for bits
- Periodic resetting is needed

🧾 Comparison Table

Strategy	Main Idea	Pros	Cons
FIFO	Oldest page out	Simple	May evict frequently used pages

Strategy	Main Idea	Pros	Cons
LRU	Least recently used page out	Effective	Needs tracking overhead
Optimal	Future use based	Best performance	Not practical
Clock	Second-chance with circular list	Efficient, approximates LRU	Less accurate than LRU
NRU	Based on reference & modify bits	Balanced and practical	Needs bit tracking

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Let me know if you'd like diagrams or examples with page reference strings and memory frames to see how each algorithm works step-by-step!

(i) Describe about the contiguous memory allocation (8 marks)
(ii) Given memory partitions of 100K, 400K, 200K, 300K and 600K. How would each of the First-fit, Best-fit, Worst-fit algorithms place processes of 112K, 317K, 212K and 62K? Which
algorithm makes the Most efficient use of memory? (8 marks)

Sure! Here's the detailed answer for both parts of your question:

(i) Contiguous Memory Allocation (8 Marks)

🧠 Definition:

Contiguous memory allocation is a **memory management technique** where **each process is allocated a single continuous block of memory** in the main memory.

🔹 Key Features:

- Entire process is loaded into **a single block** of RAM.
- Memory is divided into **fixed or variable-sized partitions**.
- Allocation is done **contiguously** (i.e., without gaps between blocks).

🔹 Types of Contiguous Allocation:

1. **Fixed Partitioning:**
   - Memory is divided into fixed-size blocks.
   - Can lead to **internal fragmentation** (wasted space inside partitions).
2. **Variable Partitioning:**
   - Memory blocks are created as per the process size.
   - Can lead to **external fragmentation** (wasted space between blocks).

🔹 Allocation Strategies:

- **First Fit:** Allocate the first hole that is big enough.
- **Best Fit:** Allocate the smallest hole that fits.
- **Worst Fit:** Allocate the largest available hole.

✅ Advantages:

- Simple and fast.
- Easy to implement.

❌ Disadvantages:

- Can lead to **fragmentation**.
- Requires **compaction** to free up space.

- Difficult to accommodate growing processes.

(ii) Memory Allocation Using Fit Algorithms (8 Marks)

📦 Given:

🔹 **Memory Partitions:**

`100K, 400K, 200K, 300K, 600K`

🔹 **Processes (in order):**

`112K, 317K, 212K, 62K`

🧮 1. First-Fit Algorithm:

- **112K** → fits in **400K** → remaining: 288K
- **317K** → next fit: **600K** → remaining: 283K
- **212K** → next fit: **300K** → remaining: 88K
- **62K** → fits in **288K** (from first allocation) → remaining: 226K

✅ **Allocation:**

- 112K → 400K
- 317K → 600K
- 212K → 300K
- 62K → remaining 288K (from 400K block)

🧮 2. Best-Fit Algorithm:

Find the **smallest partition** that fits.

- **112K** → best fit = **200K** → remaining: 88K

- **317K** → best fit = **400K** → remaining: 83K
- **212K** → best fit = **300K** → remaining: 88K
- **62K** → best fit = **88K** (from 200K block) → remaining: 26K

✅ **Allocation:**

- 112K → 200K
- 317K → 400K
- 212K → 300K
- 62K → remaining 88K (from 200K block)

🧮 3. Worst-Fit Algorithm:

Choose the **largest available partition**.

- **112K** → 600K → remaining: 488K
- **317K** → 488K → remaining: 171K
- **212K** → 400K → remaining: 188K
- **62K** → 300K → remaining: 238K

✅ **Allocation:**

- 112K → 600K
- 317K → 488K (from 600K block)
- 212K → 400K
- 62K → 300K

✅ Which is Most Efficient?

➤ Best-Fit makes the most efficient use of memory, because:

- It reduces **wasted space** (internal and external fragmentation).

- Allocates the **tightest fit** to each process.

However:

- **First-Fit** is **faster** in implementation.
- **Worst-Fit** leaves large unused gaps and is generally the **least efficient**.

Let me know if you'd like this formatted as a study sheet or in a tabular format!

Consider the following page reference string:
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1.
How many page faults would occur for the following replacement algorithms, assuming three frames? Remember all
frames are initially empty, so your first unique pages will all cost one fault each.
• FIFO replacement
• Optimal replacement
• LRU replacement

Let's solve the problem step-by-step for the **given page reference string** and **three memory frames**:

📥 Page Reference String:

```
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
```

Total references = **20**
Frames = **3**
Initially, all frames are **empty**.

🔁 1. FIFO (First-In, First-Out)

FIFO replaces the **oldest page** in memory when a page needs to be replaced.

Let's go through it:

```less
Step | Page | Frame State          | Page Fault?
-----|------|----------------------|------------
  1  |  7   | [7]                  | Yes
  2  |  0   | [7, 0]               | Yes
  3  |  1   | [7, 0, 1]            | Yes
  4  |  2   | [0, 1, 2]            | Yes (7 removed)
  5  |  0   | [0, 1, 2]            | No
  6  |  3   | [1, 2, 3]            | Yes (0 removed)
  7  |  0   | [2, 3, 0]            | Yes (1 removed)
  8  |  4   | [3, 0, 4]            | Yes (2 removed)
  9  |  2   | [0, 4, 2]            | Yes (3 removed)
 10  |  3   | [4, 2, 3]            | Yes (0 removed)
 11  |  0   | [2, 3, 0]            | Yes (4 removed)
 12  |  3   | [2, 3, 0]            | No
 13  |  2   | [2, 3, 0]            | No
 14  |  1   | [3, 0, 1]            | Yes (2 removed)
 15  |  2   | [0, 1, 2]            | Yes (3 removed)
 16  |  0   | [0, 1, 2]            | No
 17  |  1   | [0, 1, 2]            | No
 18  |  7   | [1, 2, 7]            | Yes (0 removed)
 19  |  0   | [2, 7, 0]            | Yes (1 removed)
 20  |  1   | [7, 0, 1]            | Yes (2 removed)
```

✅ **Total Page Faults (FIFO): 15**

🔮 2. Optimal Replacement

Optimal replaces the page that **will not be used for the longest time in the future**.

Going step-by-step:

1. Load 7 → fault

2. Load 0 → fault
3. Load 1 → fault
   Frames now: [7, 0, 1]
4. 2 → Replace **7** (used farthest in future) → fault
5. 0 → No fault
6. 3 → Replace **1** (used later than 0 or 2) → fault
7. 0 → No fault
8. 4 → Replace **2** → fault
9. 2 → Replace **3** (used farthest) → fault
10. 3 → Replace **4** (never used again) → fault
11. 0 → No fault
12. 3 → No fault
13. 2 → No fault
14. 1 → Replace **0** → fault
15. 2 → No fault
16. 0 → Replace **1** → fault
17. 1 → Replace **3** → fault
18. 7 → Replace **2** → fault
19. 0 → No fault
20. 1 → No fault

✅ **Total Page Faults (Optimal): 10**

🕑 3. LRU (Least Recently Used)

LRU replaces the **least recently used page**.

1. 7 → fault
2. 0 → fault

3. 1 → fault
   Frames: [7, 0, 1]
4. 2 → Replace **7** (least recently used) → fault
5. 0 → No fault
6. 3 → Replace **1** → fault
7. 0 → No fault
8. 4 → Replace **2** → fault
9. 2 → Replace **3** → fault
10. 3 → Replace **0** → fault
11. 0 → Replace **4** → fault
12. 3 → No fault
13. 2 → No fault
14. 1 → Replace **0** → fault
15. 2 → No fault
16. 0 → Replace **3** → fault
17. 1 → No fault
18. 7 → Replace **2** → fault
19. 0 → No fault
20. 1 → No fault

✅ **Total Page Faults (LRU): 12**

✅ Final Summary:

Algorithm	Page Faults
FIFO	15
Optimal	10 ✅ (Best)

Algorithm	Page Faults
LRU	12

Let me know if you'd like this in a table or with diagrams for better visualization!